



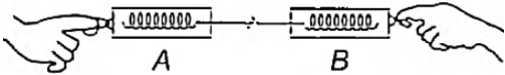
FORCE-01

Class 09 - Science

Section A

1. Bags at the top of school van are tied using a string to avoid the effect of [1]
 - a) inertia
 - b) acceleration
 - c) force
 - d) momentum
2. A mass M breaks into two pieces in the ratio 1 : 3 while at rest. If the heavier has a speed of v , the speed of the lighter is [1]
 - a) $3v$
 - b) v
 - c) $4v$
 - d) $2v$
3. If force, change in momentum and time are given by F , p and t respectively, then they are related by [1]
 - a) $F = pt$
 - b) $p = F^2t$
 - c) $F = \frac{p}{t}$
 - d) $Ft^2 = p$
4. The acceleration of a body is to be doubled from its initial value. By what factor is the acting force to be increased? [1]
 - a) half
 - b) four
 - c) two
 - d) one
5. If 10 kg and 5 N are the S.I. units of mass and force respectively, the S.I. unit of acceleration is [1]
 - a) 5 m/s^2
 - b) 2 m/s^2
 - c) 0.5 m/s^2
 - d) 1 m/s^2
6. The unit of momentum can be expressed as [1]
 - a) $\frac{N}{s}$
 - b) $N \cdot s$
 - c) $\text{kg s}^2/\text{m}$
 - d) kg m/s^2
7. When a 12 N force acts on 3 kg mass for a second, the change in velocity is (in m/s) [1]
 - a) 18
 - b) 36
 - c) 2
 - d) 4
8. A force of 5 N applied on m_1 produces an acceleration of 8 m/s^2 and when applied on m_2 produces an acceleration of 24 m/s^2 . When they are tied together, the acceleration will be [1]

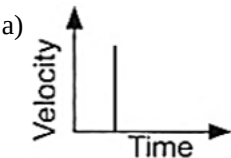
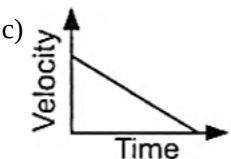
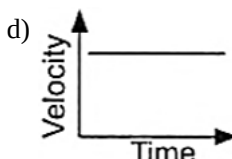
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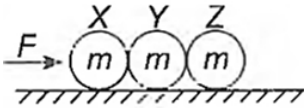
- a) act on either body at normal to each other b) always act on different bodies in opposite directions
- c) have same magnitude and direction d) always act on the same body
17. The increase in velocity acquired by a 10 kg mass in 5 seconds due to a force of 10 newton is [1]
- a) 1 m/s b) 5 m/s
- c) 10 m/s d) 500 m/s
18. A 20 kg gun fires a bullet of mass 20 g with a velocity of 400 m/s. The action on the shoulder of the person per second by the gun is: [1]
- a) 8000 N b) 8 N
- c) 4000 N d) 4 N
19. If an object experiences a net zero unbalanced force then the body: [1]
- a) moves with constant velocity b) can be accelerated
- c) cannot remain at rest d) none of these
20. Water drops sticking to the wheel come out along the tangential line due to [1]
- a) inertia b) acceleration
- c) momentum d) force
21. A water tanker filled up to $\frac{2}{3}$ rd of its height is moving with a uniform speed. On sudden application of the brake, [1] the water in the tank would
- a) be unaffected b) move backwards
- c) rise upwards d) move forwards
22. Consider two spring balances hooked as shown in the figure. We pull them in opposite directions. If the reading [1] shown by A is 1.5 N, the reading shown by B will be
- 
- a) Zero b) 2.5 N
- c) 3.0 N d) 1.5 N
23. On a 3 kg mass, 5 newton of force acts for 0.1 second. The impulse imparted to the mass is (in kg m/s): [1]
- a) 0.16 b) 1.0
- c) 1.5 d) 0.5
24. The acceleration of a body has to be doubled without any change in force. Then one has to [1]
- a) decrease the mass to half b) increase the mass by half
- c) double the mass d) cannot be made
25. According to second law of Newton, force is the cause and the outcome is: [1]
- a) time b) velocity
- c) momentum d) acceleration
26. In a rocket, fuel burns at the rate of 1 kg s^{-1} . This fuel is ejected from the rocket with a velocity of 60 km s^{-1} . [1]

This exerts a force on the rocket equal to

- a) 6000 N
- b) 600 N
- c) 60000 N
- d) 60 N

27. A motorcycle and a car are moving on a horizontal road with the same velocity. If they are brought to rest by the application of brakes, which provided equal retardation, then [1]
- a) Both will stop at the same distance
 - b) Car will stop at a shorter distance
 - c) Nothing can be predicted.
 - d) Motorcycle will stop at shorter distance
28. Which of the following statement is not correct for an object moving along a straight path in an accelerated motion? [1]
- a) Its velocity always changes
 - b) A force is always acting on it
 - c) Its speed keeps changing
 - d) It always goes away from the earth
29. The two states of motion treated alike by Newton's first law, among A, B, C, and D are [1]
A : Rest B: Uniform motion C: Uniformly accelerated. D : Non-uniformly accelerated
- a) A, D
 - b) A, C
 - c) B, C
 - d) A, B
30. Which of the following has the largest inertia? [1]
- a) your physics book
 - b) a pin
 - c) your school bag
 - d) a pen
31. In the first second of its flight, a rocket ejects $\frac{1}{40}$ of its mass with a velocity of 1200 m s^{-1} . The acceleration of the rocket is [1]
- a) 40 m s^{-2}
 - b) 60 m s^{-2}
 - c) 30 m s^{-2}
 - d) 20 m s^{-2}
32. The one which has the least inertia among the following: 1 kg stone, 2 kg ball, a train compartment and a cup of tea is [1]
- a) 2 kg ball
 - b) 1 kg stone
 - c) a train compartment
 - d) a cup of tea
33. A passenger in a moving train tosses a coin which falls behind him. It means that motion of the train is [1]
- a) along circular tracks
 - b) uniform
 - c) accelerated
 - d) retarded
34. The minimum number of unequal forces that can make zero resultant is [1]
- a) ten
 - b) four
 - c) three
 - d) two
35. A bullet of mass 10 g is fired from a rifle. The bullet takes 0.003 second to move through its barrel and leaves it with a velocity of 300 m/s. What is the force exerted on the bullet by the rifle? [1]
- a) 250N
 - b) 1000 N

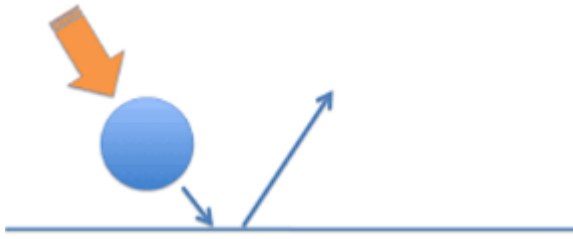
- c) 400N
d) 200N
36. When no external force acts on an object, the physical quantity that remains conserved is [1]
a) force
b) momentum
c) acceleration
d) velocity
37. Action reaction forces act: [1]
a) on the same body
b) on different bodies
c) in the same direction
d) along different bodies
38. Fireman holds a hose by exerting a _____. [1]
a) momentum
b) friction
c) force
d) an acceleration
39. A force of 5 N gives a mass m_1 an acceleration of 10ms^{-2} and mass m_2 an acceleration of 20ms^{-2} . What acceleration would if give it both the masses were tied together? [1]
a) 7.65ms^{-2}
b) 7.00ms^{-2}
c) None of these
d) 6.67ms^{-2}
40. The SI unit of force is: [1]
a) Newton
b) Newton per second
c) Newton per square metre
d) Newton metre
41. A player caught a cricket ball of mass 200 g moving at a rate 60 m/s. If the catching process completes in 0.4 s, the force of the blow exerted by the ball on the hand of the player is equal to [1]
a) 300 N
b) 30 N
c) 150 N
d) 3 N
42. A plate, a ball and child all have the same mass. The one having more inertia is the [1]
a) child
b) plate
c) All have equal inertia
d) ball
43. Which of the following is the velocity-time graph of a moving particle on which net external force is zero? [1]
a) 
b) 
c) 
d) 
44. The change in the momentum of a body in 0.01 seconds is 10kg ms^{-1} . The force acting on this body is [1]
a) 100 N
b) 0.1 N
c) 10 N
d) 1000 N
45. A force can be completely described by: [1]

- a) its magnitude
b) neither magnitude nor direction
c) its magnitude and direction
d) its direction
46. Two blocks made of different metals identical in shape and size are acted upon by equal forces which cause them to slide on a horizontal surface. The acceleration of the second block is found to 10 times that of the first. What is the ratio of the mass of the second to the first? [1]
- a) 1 : 5
b) 1 : 1
c) 1 : 10
d) 10 : 1
47. A cup of water holding a steel ball (m) is allowed to fall under gravity. The reaction between the ball and the base of the cup is [1]
- a) $\frac{mg}{2}$
b) zero
c) mg
d) 2mg
48. Three identical bodies each of mass m, are pushed by a force F on a frictionless table as shown in figure. [1]
- 
- a) Net force on X is F.
b) Net force on Y is equal to zero.
c) Force applied by Y on Z is equal to net force on X.
d) Force applied by X on Y is $\frac{F}{3}$.
49. An object of mass 2 kg is sliding with a constant velocity of 4ms^{-1} on a frictionless horizontal table. The force required to keep the object moving with the same velocity is [1]
- a) 32 N
b) 2 N
c) 0 N
d) 8 N
50. While catching a stone thrown by your friend you pull the hands back to [1]
- a) avoid the breaking of the stone
b) avoid getting hurt
c) increase the time to slow down
d) decrease the time to slow down
51. Impulse has the S.I. unit of _____ [1]
- a) newton
b) N-s
c) joule
d) m/s^2
52. A goalkeeper in a game of football pulls his hands backwards after holding the ball shot at the goal. This enables the goalkeeper to [1]
- a) reduce the force exerted by the ball on hands
b) exert larger force on the ball
c) increase the rate of change of momentum
d) decrease the rate of change of momentum
53. A truck carries sand. Due to hole in the base of truck, significant amount of sand falls continuously. For the same condition of working of the engine, the speed of the truck should [1]
- a) increase
b) cannot be found
c) remain the same
d) decrease

Section B

54. What does the given image illustrate?

[1]



a) Second law of motion

b) Law of conservation of momentum

c) Third law of motion

d) First law of motion

55. Which effect of force is observed in the given image?

[1]



a) A force can change the speed of a moving body

b) A force can change the shape of the object

c) A force can move a stationary body

d) A force can stop a moving body

56. The given image correlates with which law of motion?

[1]



a) First law of motion

b) Second law of motion

c) Law of momentum

d) Third law of motion

57. In the given below image, the goalkeeper held his palm in the path of the moving ball to show that

[1]



a) A force can change the speed of a moving body

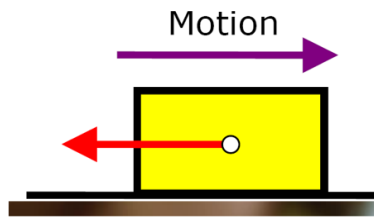
b) A force can move a stationary body

c) A force can stop a moving body

d) A force can change the shape of the object

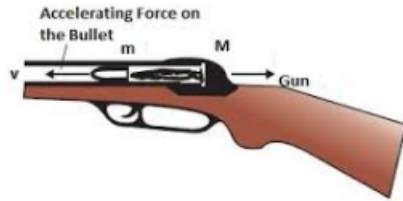
58. The image shows a force which always opposes the motion of one body over another body. This is:

[1]



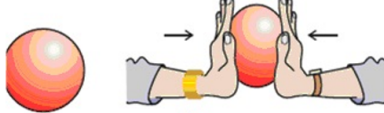
- a) Gravitational force
- b) Frictional force
- c) Electrostatic force
- d) Magnetic force

59. The image shows the firing of a gun. What happens when a gun is fired? [1]



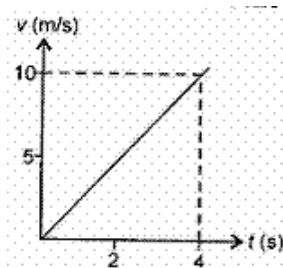
- a) Exerts a backward force on the bullet
- b) Exerts a forward force on the arm
- c) Exerts a forward force on the bullet
- d) None of these

60. When we press a rubber ball with both the hands, it [1]



- a) becomes light
- b) None of these
- c) changes composition
- d) changes its shape

61. The v-t graph of a body of 5 kg moving with the help of a force is shown. Then the force involved is [1]



- a) 12.5 N
- b) 2.0 N
- c) 20 N
- d) 125 N

62. The given image shows an effect of force that it can [1]



- a) Change the shape of the object
- b) Move a stationary object
- c) Stop a moving object
- d) Change the size of the object

63. As the sailor jumps forward out of a rowing boat, the force on the boat moves it backward. This is an example of [1]



- a) law of conservation of momentum
- b) second law of motion
- c) first law of motion
- d) third law of motion

Section C

64. A gun recoiled to _____ the momentum. [1]

- a) decrease
- b) conserve
- c) increase
- d) change

65. Inertia is of _____ types. [1]

- a) three
- b) four
- c) two
- d) one

66. Balanced forces may _____ a body. [1]

- a) Accelerated
- b) Move
- c) Retard
- d) Deform

67. Newton's first law gives the _____ definition of force. [1]

- a) quality
- b) quantitative
- c) qualitative
- d) None of these

68. Newton's third law tells that _____ force does not exist. [1]

- a) motion
- b) isolated
- c) movement
- d) momentum

69. The total _____ contained in a body is called momentum. [1]

- a) none of these
- b) momentum
- c) fat
- d) motion

70. The rate of change of momentum is _____ to the applied force. [1]

- a) indirect proportional
- b) correct proportional
- c) directly proportional
- d) None of these

71. A high jumper runs for a while before taking high jump so that the inertia of _____ help him in taking a long jump. [1]

- a) shape
- b) direction
- c) rest
- d) motion

72. Leaves get detached from a tree due to _____. [1]

- a) conservation of momentum
- b) change in momentum

c) acceleration

d) inertia

73. If no external force acts on a system of particles then the _____ of the system remains conserved.

[1]

a) movement

b) inertia

c) momentum

d) motion